

# Ghilmo Choi

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## Education

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| <b>Korea Aerospace University</b> , Republic of Korea<br>B.S. in Mechanical Engineering | Mar 2021 – Feb 2027<br>(expected) |
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- Official GPA: **4.00 / 4.5** (Major GPA: **4.08 / 4.5**)
- Expected GPA after current-semester grades are posted: **4.08 / 4.5** (Major GPA: **4.18 / 4.5**)
- Main Coursework: Introduction to Machine Learning, Automatic Control, Mechanical Vibrations, Data Structures and Algorithms, Probabilistic Robotics

## Research Interests

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- On-device vision-language models for natural-language robotic mission execution
  - Perception, Prediction, and Decision Making for Autonomous Driving
  - Machine learning, deep learning, and AI model architectures for robotic intelligence

## Research / Project Experience

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| <b>ASOL</b> (Autonomous Systems and Optimization Lab)<br>- <i>Undergraduate Researcher</i> | Korea Aerospace Univ.<br>June 2025 – Present |
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- On-device VLM-based robotic mission execution**  
Developed a natural-language mission execution framework that converts user commands into executable robot behaviors on a UGV platform
- Hallucination-aware and SLAM-grounded task planning**  
Designed an Actor–Guard–Critic planning loop and integrated RTAB-Map, object detection, structured world states, and ROS2/Nav2 control for real-robot language-to-motion execution
- AirSim-based perception, reconstruction, and path generation**  
Extending the framework with YOLOE, NanoOWL, and Grounding DINO for object detection on AirSim data, NVBlox for 3D reconstruction, and OMPL for path generation

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| <b>KAUVOY</b> Autonomous Driving Club<br>- <i>Localization Part</i> | Korea Aerospace Univ.<br>Nov 2024 – Nov 2025 |
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- EKF-based localization**  
Implemented Dual EKF localization on an ERP42 autonomous vehicle by fusing IMU, wheel encoder, and RTK-GPS measurements
- Autonomous parking and path tracking**  
Developed and tested a real-vehicle parking pipeline covering cone perception, parking-space decision, Hybrid A\* path planning, and Pure Pursuit-based path tracking

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| - <i>Team Leader</i> | Nov 2025 – present |
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- Finite-State-Machine-based decision-making module for autonomous driving scenarios**

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| Course Project, Introduction to Machine Learning<br>- <b>End-to-End Machine Learning Classification and Regression Pipeline</b> | Korea Aerospace Univ.<br>Aug 2025 – Nov 2025 |
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- Built and validated classification and regression pipelines using preprocessing, dimensionality reduction, ensemble methods, and FCNNs; achieved **0.915 macro F1** on a 4-class imbalanced classification task

## Honors & Activities

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- Finalist, **2025 College Mobility Innovation Competition**
  - Excellence Award, **Drone X AI Marketing Challenge** (2026)
  - 2026 - **Airbus 101 Competition** (5th Place)

## Publications

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- **Co-author**, "*On-Device Vision Language Model Based Natural Language Mission Execution Framework*," submitted to the International Conference on Control, Automation and Information Sciences (**ICCAIS**), 2025
- **Co-author**, "*ROS `robot_localization` Dual EKF: Verification of Localization Accuracy and Application to Autonomous Driving Path Tracking*," submitted to the Korean Society of Automotive Engineers (**KSAE**), 2025

## Technical Skills

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- Python, C/C++, MATLAB, Linux(Ubuntu), ROS1, ROS2, Docker, Git